

Partial Products, Clear Thinking



Grade 4 • Mathematics
NC.4.NBT.5

Connect area models with partial products.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

The library team packs 182 cards in each of 3 boxes. Find the total. Show partial products and explain why they can be added.

$(180 \times 3) + (2 \times 3) = 540 + 6 = 546$. The parts account for every box or card exactly once.



Build a second model together

The reading group packs 341 cards in each of 3 boxes. Find the total. Show partial products and explain why they can be added.

Discuss: What should the first step find?



Your turn • 1

The music club packs 27 cards in each of 29 boxes. Find the total. Show partial products and explain why they can be added.

Represent → solve → explain → check



Your turn • 2

The design team packs 16 cards in each of 22 boxes. Find the total. Show partial products and explain why they can be added.

Represent → solve → explain → check



Your turn • 3

The school garden packs 248 cards in each of 3 boxes. Find the total. Show partial products and explain why they can be added.

Represent → solve → explain → check



Your turn • 4

The science club packs 233 cards in each of 4 boxes. Find the total. Show partial products and explain why they can be added.

Represent → solve → explain → check



Exit question • Explain independently

The library team packs 191 cards in each of 4 boxes. Find the total. Show partial products and explain why they can be added.

Include a check. What would you teach someone else?



Teacher review and next steps

1. 783

Exit: 764

2. 352

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

3. 744

4. 932

